Stochastic Processes 2021/22 Semester 1

Tutorial 6

1. Let $X_1 \sim B(n_1, p), X_2 \sim B(n_2, p), X_3 \sim B(n_3, p)$, where $p \in (0, 1)$, be three mutually independent random variables. By thinking in terms of coin tosses, explain why

$$X_1 + X_2 + X_3 \sim B(n_1 + n_2 + n_3, p).$$

2. For each n, let $X_n \sim B(n, p_n)$ where p_n is such that as $n \to \infty$, $p_n \to 0$ and $np_n \to 0$. Show that

$$\lim_{n\to\infty} \mathbf{P}(X_n = 0) = 1.$$

3. Let $N \sim \Pi(\lambda)$, $\lambda > 0$. Assume that (a random number) N balls are taken and that, independently of everything else, each of them is coloured "red" with probability p, "white" with probability q, and "yellow" with probability r (where $p, q, r > 0$ and $p + q + r = 1$). Let R be the total number of reds, W of whites and Y of yellows. Show that

   (i) $R \sim \Pi(\lambda p)$; $W \sim \Pi(\lambda q)$; $Y \sim \Pi(\lambda r)$;

   (ii) the three random variables R, W, Y are mutually independent.

4. Let X_1, X_2, X_3 be three i.i.d random variables with common distribution $G(p)$, where $p \in (0, 1)$. For $1 \leq k \leq l - 2$, find

$$\mathbf{P}(X_1 = k \mid X_1 + X_2 + X_3 = l).$$

5. Let $Y_1 \sim Exp(\alpha_1)$ and $Y_2 \sim Exp(\alpha_2)$ be independent random variables. Find the distribution functions of the random variables

   (i) $Z = \max(Y_1, Y_2)$ and

   (ii) $T = \max(Y_1, Y_2) - \min(Y_1, Y_2)$.

6. Prove that if $X \sim Gamma(\alpha, n)$, then $\mathbf{E}(X) = n/\alpha$ and $\mathbf{Var}(X) = n/\alpha^2$.

7. Calls arrive at a switchboard as a Poisson process with rate $\lambda = 4$ per minute. Find the probabilities that, with respect to any fixed origin in time, the twelfth subsequent call arrives:

   (a) before two minutes have elapsed [*hint: consider the distribution of the number of calls arriving in these two minutes*],

   (b) after three minutes and before five minutes have elapsed.

8. Let $N(\cdot)$ be a Poisson process with (constant) rate λ on $\mathbb{R}_+$.

   (a) For any $t > 0$, find the coefficient of variation (standard deviation divided by mean) of $N[0, t]$ and comment.

   (b) For any $s, t > 0$, find the distribution of $N[0, t]$ conditional on the event $N[0, t + s] = n$. [*Hint: use the "conditional uniform" property of the Poisson process.*]

9. From time $t = 0$ passengers arrive at a railway station as a Poisson process with rate λ for a train which leaves at time $t = s$. Show that the sum of the times passengers have to wait at the station until the train leaves has expected value $\lambda s^2/2$. [*Hint: use the 'conditional uniform' property to find first the expected value conditional on a total of n passengers arriving in* $[0, s]$*; then find the unconditional expectation.*]